GANDHIMATHI (MATHI) PADMANABAN

University of Michigan-Dearborn gmathi@umich.edu - Website - GitHub - LinkedIn

RESEARCH PROFILE

I build and evaluate adaptable AI systems that are driven and constrained by physical measurements from multi-modal data in safety-critical settings involving human interaction. My research sits at the intersection of computer vision, machine learning, and human-centered automation, applied to transportation safety. My long-term research agenda is to make these systems auditable, so that the people who share roads and other spaces with them are not the ones who discover when they are wrong.

EDUCATION

Aug 2026 **Ph.D., Industrial and Systems Engineering** - *University of Michigan-Dearborn, USA*
Dissertation: "Adaptable Machine Learning and Computer Vision Frameworks for Road Safety: Applications to Driver Behavior and Driver-Bicyclist Interaction Research"
Advisor: Dr. Fred Feng

Aug 2021 **M.S., Human Centered Design and Engineering** - *University of Michigan-Dearborn, USA*
Thesis: "Computational Human Performance Modeling using Queuing Network in an Open-Source Platform"
Advisor: Dr. Fred Feng

Apr 2013 **B.E., Computer Science and Engineering** - *Anna University, India*
Thesis: "Automated Detection of Modifications in Software Requirement Traceability Links"
Advisor: Prof. Ramachandran Alagarsamy

Certifications: Rackham DEI Certificate (2025); PG Diploma in Computer Applications (2011)

RESEARCH EXPERIENCE

2022-2026 **Graduate Student Research Assistant** - *University of Michigan-Dearborn, USA*
Advisor/PI: Dr. Fred Feng | Funding: National Science Foundation (award 2142757), Ford Motor Company

2011-2013 **Research Assistant** - *Anna University, India*
Advisor/PI: Prof. Ramachandran Alagarsamy

PUBLICATIONS

Peer-Reviewed Conference Papers

1. **Gandhimathi Padmanaban**, Fred Feng, Edward Dai, Ankit Saini, Guopeng Hu, and Yanan Zhao. "A Comparative Analysis of Acceleration and Deceleration Profiles for Aggressive Driving Styles and Fuel Economy Test Cycles". *WCX SAE World Congress Experience*. 2025. DOI: http://dx.doi.org/10.4271/2025-01-8605.
2. Rayane Moustafa, **Gandhimathi Padmanaban**, and Fred Feng. "Quantifying Drivers-Overtaking-Bicyclists with Surrogate Safety Measures Derived from High-Resolution Digital Lidar". *TRB Annual Meeting*. 2026. URL: https://par.nsf.gov/biblio/10679666.
3. **Gandhimathi Padmanaban**, Nathaniel P. Jachim, Hala Shandi, Lilit Avetisyan, Garrett Smith, Howrah Hammoud, and Feng Zhou. "An Autonomous Driving System - Dedicated Vehicle for People with ASD and their Caregivers". *AutomotiveUI '21 Adjunct: 13th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*. Association for Computing Machinery, 2021, pp. 142-147. DOI: https://doi.org/10.1145/3473682.3480282.

Papers Under Review

1. **Gandhimathi Padmanaban**, Rayane Moustafa, and Fred Feng. "A Geometry-Informed Computer Vision Method for Detecting and Examining Overtaking Vehicles From A Bicycle". *Submitted, IEEE Transactions on Intelligent Transportation Systems*. URL: https://par.nsf.gov/biblio/10679667.
2. **Gandhimathi Padmanaban**, Fred Feng, Edward Dai, Ankit Saini, Guopeng Hu, and Yanan Zhao. "A Machine Learning Framework to Identify Aggressive Driving Based on Vehicle Kinematics and Driver Pedal Operations". *Abstract accepted, SAE WCX 2027.*

Working Papers

1. **Gandhimathi Padmanaban** and Fred Feng. "Vision-Based Lateral Passing Distance Estimation from Bicycle-Mounted Cameras: A Projective Geometry Approach". *In preparation.*
2. **Gandhimathi Padmanaban** and Fred Feng. "A Multi-View Vehicle Image Dataset and Two-Stage Pipeline for Fine-Grained Vehicle-Type Recognition at Ground Level". *In preparation for CVPR 2027.*

PROFESSIONAL EXPERIENCE (INDUSTRY)

2017-18	**Consultant / Lead Front End Developer**, Deloitte (Offices of the US), Bangalore, India
2015-17	**Programmer Analyst**, Cognizant, Chennai, India
2013-15	**Software Engineer Level-II**, Syncfusion, Chennai, India

TEACHING AND MENTORING

2026	**Teaching Assistant** - *Python as an Engineering Tool, University of Michigan-Dearborn* Assisted students during lessons and graded final projects.
2024-*Now*	**Certified Instructor** - *The Carpentries* Hands-on computational methods workshops (Python data analysis and image processing, Git version control, Unix shell, SQL, OpenRefine) for researchers in biosciences, library science, and data-intensive fields; five as instructor. Maintainer of the Library Carpentry computational thinking lesson; contributor to AI Carpentry curriculum development. CDC (2026) | Stanford (2026) | U. Tennessee Knoxville (2025) | CZI (2024) | U. Michigan (2024)
2023-*Now*	**Instructor & Mentor** - *WoCCode - University of Michigan-Ann Arbor* Machine Learning Workshop (Feb 2024) | Summer 2023 Boot Camp
2010-12	**Student Instructor / Teaching Assistant** - *Anna University, India* Courses: Artificial Intelligence, Probability and Queuing Theory, Transforms and PDEs, Systems Software Laboratory

TECHNICAL EXPERTISE

AI/ML Methods	Geometric plausibility checks on model output; geometry-informed computer vision; deep learning (CNN, RNN, Transformers, ViT); supervised learning and ensembles; sequential and dynamical system modeling; Bayesian methods; out-of-distribution robustness and cross-site generalization; conformal and selective prediction (in progress)
Perception & Sensing	Multi-object detection and tracking (RT-DETR, ByteTrack, YOLOv5); multimodal camera and LiDAR pipelines; sensor fusion; vehicle kinematics modeling; ROS; Vision-Language Models
Evaluation, Data & Standards	Evaluation and measurement validity; benchmark and test-procedure critique; dataset construction, annotation, documentation and public release; cross-platform evaluation
Human Systems & HCAI	Human performance modeling (queueing networks); human-in-the-loop handoff design; abstention and selective prediction so a system can defer when unsure; human-centered automation on the machine side; decision-support evaluation
Software & Platforms	**Primary**: Python (PyTorch, TensorFlow, scikit-learn, OpenCV, pandas), Git, Docker, LaTeX; **Working**: Julia, R, MATLAB, C#, SQL, UI frameworks, full-stack web; **HPC**: Great Lakes cluster (Slurm); production software delivery in industry
Foundations	Optimization; statistical learning; probability and queueing theory; multivariable calculus; mathematical modeling; projective geometry; dynamical systems
Research Practice	Experimental design; statistical analysis; scientific writing; peer review; IRB-reviewed human-subjects studies (surveys, interviews, usability testing); cross-validation; research project management and stakeholder communication
Domains	Human-centered automation; safety-critical and decision-support AI; autonomous and connected vehicles; transportation safety; naturalistic driving studies; vulnerable road user safety; human factors

HONORS AND AWARDS

- Student Visionary Award - International Forum on Research Excellence (IFoRE'25), Sigma Xi, 2025
- Upsilon Pi Epsilon (UPE) Scholarship - Awarded for exceptional academic performance, extracurricular involvement, and leadership within the computing community, 2024

- Global Finalist - NASA Space Apps Challenge, 2023
- Irma M. Wyman Scholar, *Center for the Education of Women (CEW+), University of Michigan*, 2020-2021 ($11,500)
- Non-Resident Graduate Student Scholar, *University of Michigan-Dearborn*, 2020-2021 ($13,000)
- Deloitte Hackathon Special Mention, 2017 | Syncfusion Hackathon 2nd Place, 2015 (INR 35,000)
- To My Younger Self Mentorship Program - Selected mentee, INFORMS Women in OR/MS (WORMS), 2026

TALKS AND PRESENTATIONS

Talks

- **WoCCode, University of Michigan Ann Arbor**, Feb 2024
  NASA SpaceApps Experience: DigitwiML, an Open-Source Project to model Digital Twin of C.elegans in Space
- **University of Michigan-Dearborn**, Sep 2021
  Guest Talk: IMSE 501 Human Factors & Ergonomics

Conference Presentations

- **International Cycling Safety Conference (ICSC)**, Nov 2026 *(Poster, accepted)*
  Vision-Based Lateral Passing Distance Estimation from Bicycle-Mounted Cameras: A Projective Geometry Approach
- **Transportation Research Board Annual Meeting**, Jan 2026 *(Poster)*
  A Geometry-Informed Computer Vision Method for Detecting and Examining Overtaking Vehicles From A Bicycle
- **Transportation Research Board Annual Meeting**, Jan 2026 *(Presentation)*
  Quantifying Drivers-Overtaking-Bicyclists with Surrogate Safety Measures Derived from High-Resolution Digital Lidar *(Moustafa, Padmanaban, Feng)*
- **International Forum on Research Excellence (IFoRE'25)**, Oct 2025 *(Workshop Session)*
  Hybrid Physics-Data Modeling for Sustainable Transportation: Bridging Classical Models and Modern AI
- **WCX SAE World Congress Experience**, Apr 2025 *(Paper Presentation)*
  A Comparative Analysis of Acceleration and Deceleration Profiles for Aggressive Driving Styles and Fuel Economy Test Cycles
- **Automotive User Interfaces (AutoUI) Conference**, Sep 2021 *(Paper Presentation)*
  An Autonomous Driving System: Dedicated Vehicle for People with ASD and their Caregivers

RESEARCH SOFTWARE & OPEN SOURCE PROJECTS

- **Geometry-informed overtaking tracker** - Automated measurement of vehicle-bicyclist interactions with a physical plausibility check on detector output (open-source, Python) - GitHub
- **Vehicle Type Classifier** - Injury-risk vehicle classification that abstains under low confidence (open-source, Python) - GitHub
- **DigitwiML** (NASA SpaceApps Global Finalist, 2023): Machine learning platform for digital twin modeling of biological dynamics in space environments - GitHub

SERVICE AND LEADERSHIP

Leadership: President, Upsilon Pi Epsilon Michigan Beta Chapter (2024-2025)

Memberships: SAE International (AI in Simulation Task Force), Sigma Xi, Alpha Pi Mu, Society for Industrial and Applied Mathematics, ACM, HFES, INFORMS

Reviewer: IEEE T-ITS, TRBAM 2026, ICIS 2025, AutoUI 2024, CHI 2024, AMCIS 2024, CUI 2024, IMX 2024, DIS 2024

Community: MIDAS AI Summit Student Volunteer (2023, 2025) • WoC|CodeNova Hackathon Judge (2025) • Hackathon mentor (WoCCode, UPE) • Ph.D. orientation panelist • WoCCode Mentor • Carpentries Certified Instructor & Lesson Maintainer • NASA SpaceApps & MHacks Judge